

1.2 Initial-Value Problems

As we saw in section 1.1, differential equations can have more than one solution. For a 1st order DE, the general solution includes an arbitrary constant C . One particular solution corresponds to each value of C .

For example, the general solution of

$$\frac{dy}{dx} = 2xy \quad \text{is} \quad y = Ce^{x^2}.$$

To determine C , we need one more piece of information about y .

For instance, if we know that $y(0) = 5$, then $C = 5$ so that $y = 5e^{x^2}$.

An initial-value problem (IVP)

Consists of : $\begin{cases} 1. \text{ A DE} \\ 2. \text{ Initial conditions.} \end{cases}$

If we have an n^{th} -order DE, the IVP will have n initial conditions.

1st order IVP

$$\begin{cases} y' = f(x, y) \leftarrow \text{DE} \\ y(x_0) = y_0 \leftarrow \text{Initial Condition} \end{cases}$$

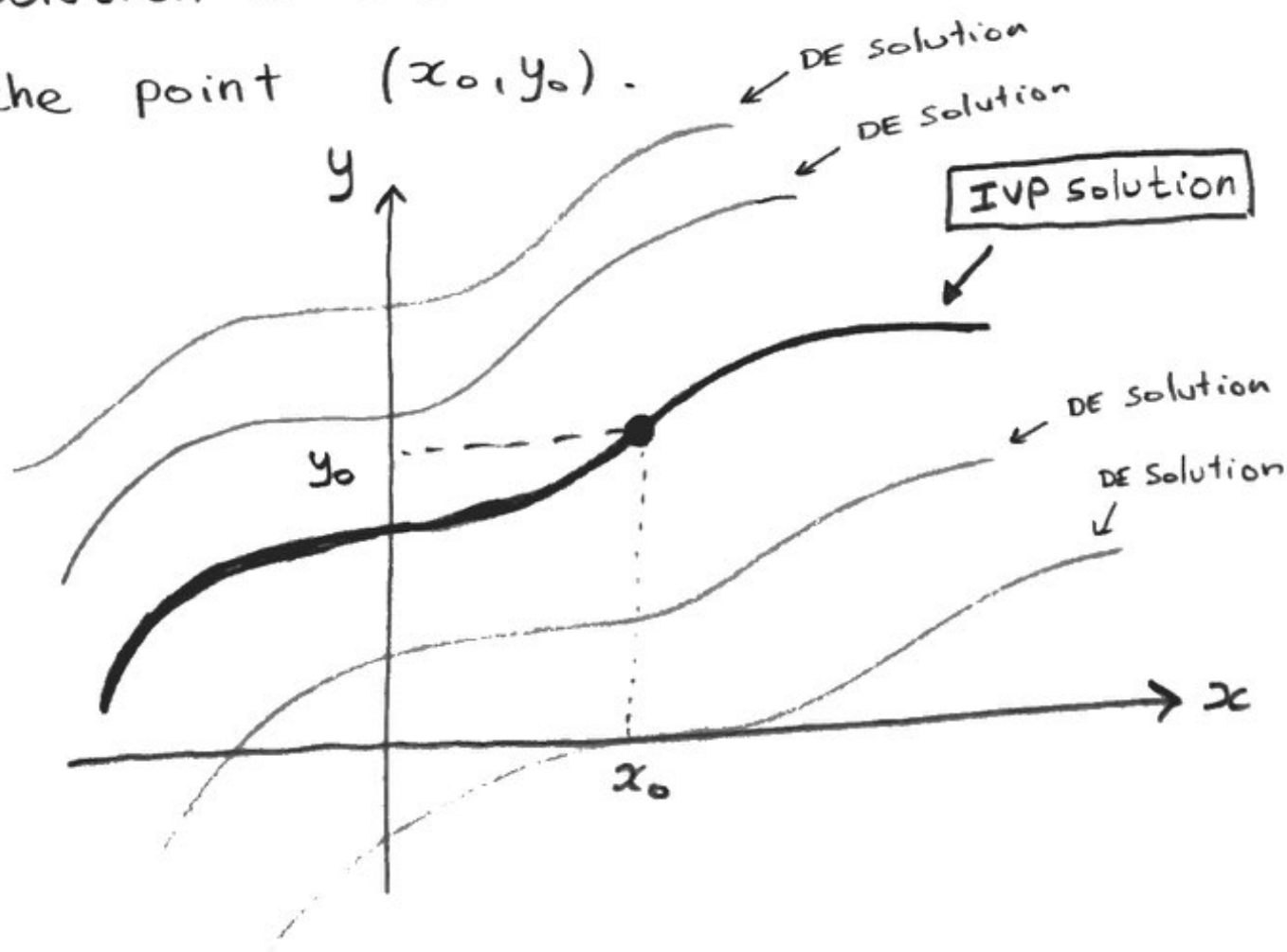
2nd order IVP

$$\begin{cases} y'' = f(x, y, y') \leftarrow \text{DE} \\ y(x_0) = y_0 \\ y'(x_0) = y_1 \end{cases} \leftarrow \text{Initial Conditions}$$

Geometric interpretation

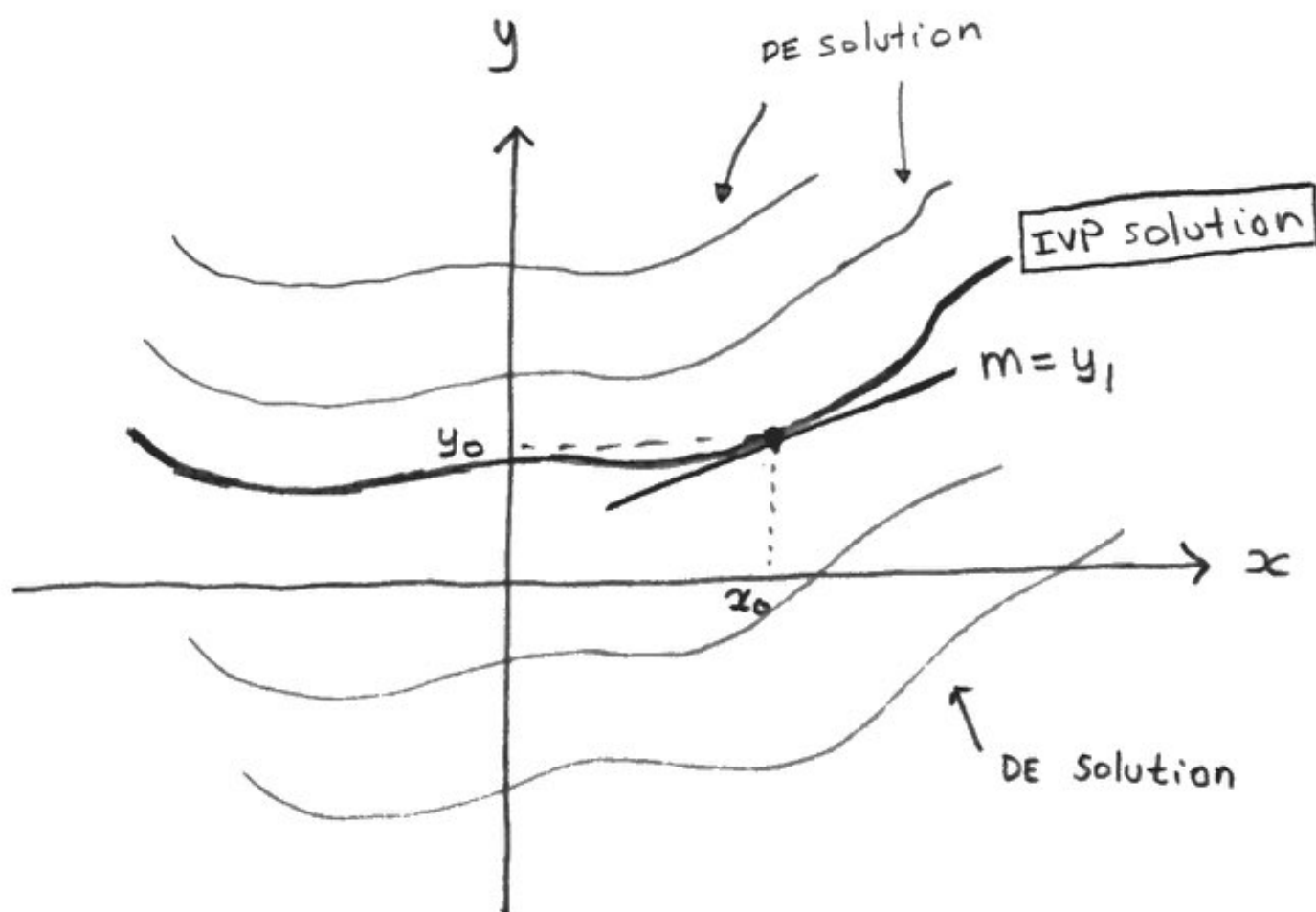
$$\begin{cases} 1^{\text{st}} \text{ order} \\ \text{IVP} \end{cases} \begin{cases} y' = f(x, y) \leftarrow \text{DE} \\ y(x_0) = y_0 \end{cases}$$

To solve the IVP, we are looking for a solution of the DE that passes through the point (x_0, y_0) .



$$\begin{array}{l} 2^{\text{nd}} \text{ order} \\ \text{IVP} \end{array} \left\{ \begin{array}{l} y'' = f(x, y, y') \leftarrow \text{DE} \\ y(x_0) = y_0 \\ y'(x_0) = y_1 \end{array} \right.$$

To solve the IVP, we are looking for a solution of the DE that passes through point (x_0, y_0) with a slope of y_1 .



The IVPs that we will encounter in this course will always have a unique solution.

For the IVP
$$\begin{cases} y' = f(x, y) \\ y(x_0) = y_0 \end{cases},$$

if the functions $f(x, y)$ and $\frac{\partial f}{\partial y}$

are continuous, the IVP has a

unique solution. In chapter 4, we

will discuss uniqueness of solutions of

2nd-order IVPs.